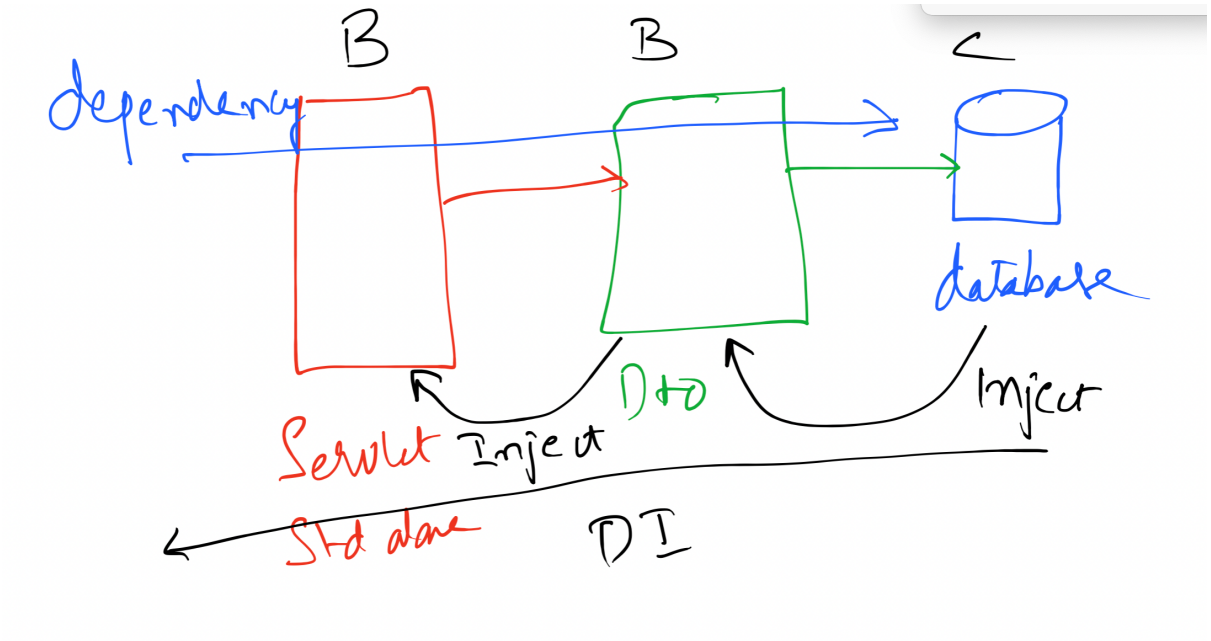
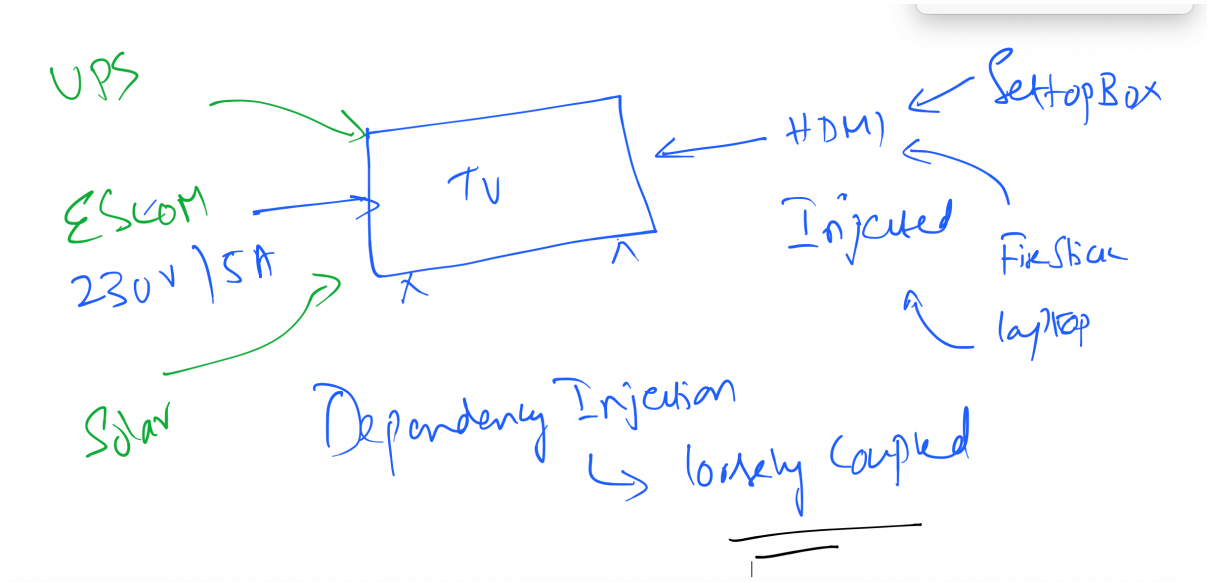
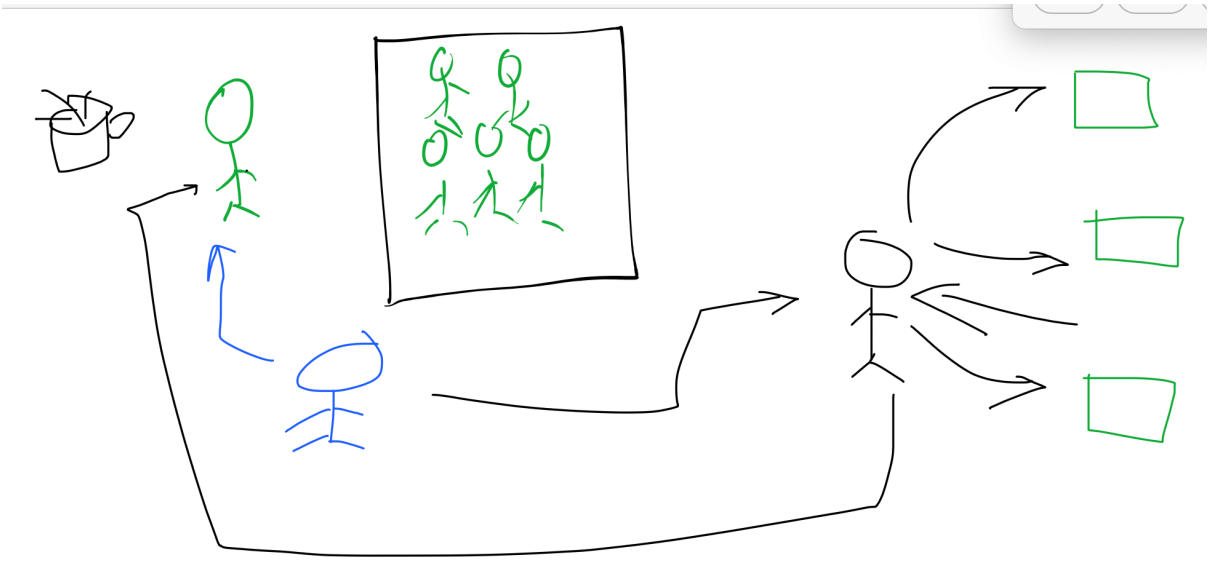
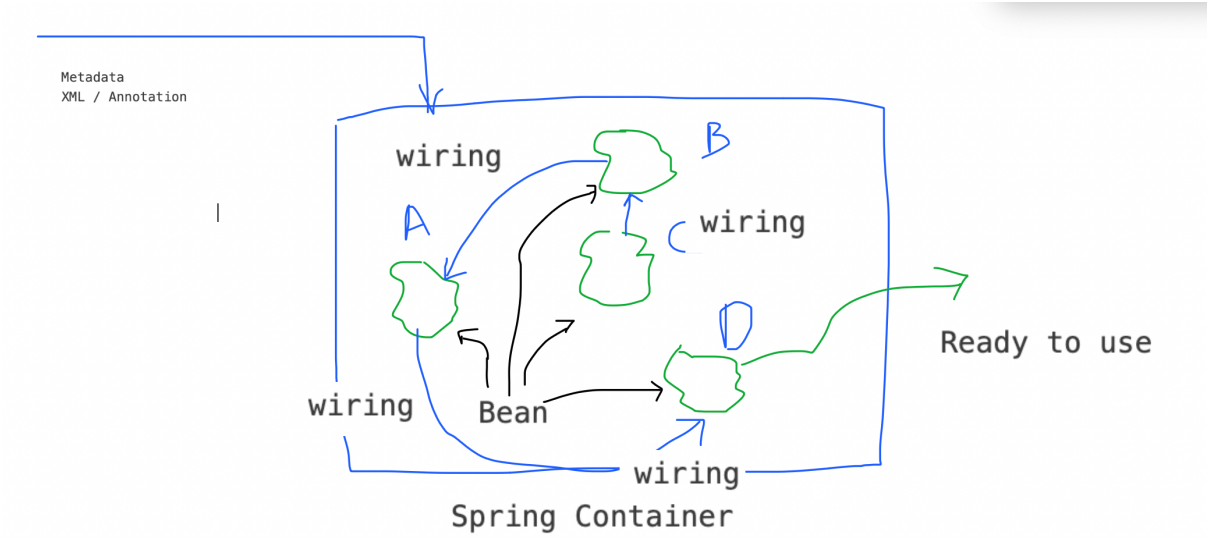

B
B
C
dependency
database
Servlet
Inject
Dto
Inject
Std alone
DI


UPS
ESKOM
230V/5A
Solar
TV
HDMI
SettopBox
Injected
FireStick
laptop
Dependency Injection
loosely coupled



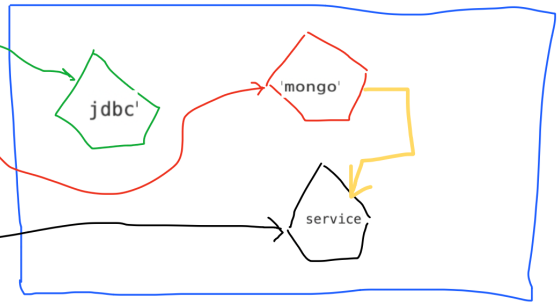


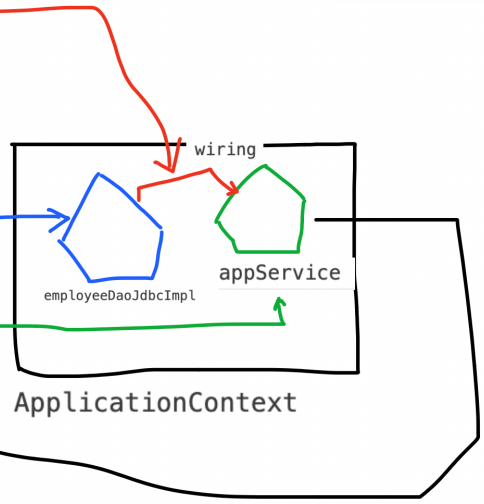

```
interface EmployeeDao {
    void addEmployee();
}

public class EmployeeDaoJdbcImpl implements EmployeeDao {
    public void addEmployee() {
        // logic to insert into RDBMS
    }
}

public class EmployeeDaoMongoDbImpl implements EmployeeDao {
    public void addEmployee() {
        // logic to insert into MongoDB --> NoSQL
    }
}
```

```
public class AppService {
    private EmployeeDao empDao;

    public void setEmpDao(EmployeeDao dao) {
        this.empDao = dao;
    }

    public void insert() {
        this.empDao.addEmployee();
    }
}
```

```
<beans>
        <bean id="jdbc" class="pkg.EmployeeDaoJdbcImpl" />
        <bean id="mongo" class="pkg.EmployeeDaoMongoDbImpl" />
        <bean id="service" class="pkg.AppService">
            <property name="empDao" ref="mongo" />
        </bean>
</beans>
```

```
service.setEmpDao(mongo);
```



ApplicationContext

```java
interface EmployeeDao {
        void addEmployee();
}

@Repository
public class EmployeeDaoJdbcImpl implements EmployeeDao {
        public void addEmployee() {
            // logic to insert into RDBMS
        }
}

@Service
public class AppService {
        @Autowired
        private EmployeeDao empDao;

        public void insert() {
            this.empDao.addEmployee();
        }
    }

ApplicationContext ctx = new AnnotationConfigApplicationContext();
ctx.scan("com.cisco.prj");
ctx.refresh();

AppService ser = ctx.getBean("appService", AppService.class);
ser.insert();
```